Should the tactic be scaled when the experiment lower bound is tested ag

Estimand

Incremental conversions of the tactic. Scale only if the experiment lower bound is positive and test cost divided by that bound is within the planning cap. The revenue model and Markov do not set the budget.

Known effect on this generator: 240 incremental conversions.

Estimates

- geo_did: 284 (interval 102 to 482).
- ols: 463 (interval 189 to 738).
- markov_mta: 1560 (interval 1560 to 1560).

The generator also places the tactic on 2% of baseline journeys. Those journeys are not incremental conversions, so the Markov removal effect is not the planted effect. Markov returns a point, not an interval. Agreement with it is not claimed.

Test cost 12.
Planning cap 0.10 per incremental conversion.

Recommendation

The intervals disagree. The experiment drives the go or no-go. Do not scale. Cost per incremental conversion at the experiment lower bound is 0.1178, above the cap 0.1000. The models stay diagnostic.

Path log without the tactic

Same panel, with the tactic removed from the path log. Geo and the revenue model are unchanged. Markov: 0 (interval 0 to 0). The intervals disagree. The experiment drives the go or no-go. Do not scale. Cost per incremental conversion at the experiment lower bound is 0.1178, above the cap 0.1000. The models stay diagnostic. This read does not replace the recommendation.

Operating characteristics

Fresh draws of this generator, 300 seeds per row, the same experiment interval and the same cap test. The planted effect is set so the true cost is the row value.

- true cost 0.03: scale in 86%, the interval covers the truth in 93% (Monte Carlo standard error 1.5%).
- true cost 0.05: scale in 29%, the interval covers the truth in 93% (Monte Carlo standard error 1.5%).
- true cost 0.08: scale in 6%, the interval covers the truth in 93% (Monte Carlo standard error 1.5%).
- true cost 0.10: scale in 3%, the interval covers the truth in 93% (Monte Carlo standard error 1.5%).

- true cost 0.12: scale in 1%, the interval covers the truth in 93% (Monte Carlo standard error 1.5%).
- true cost 0.15: scale in 1%, the interval covers the truth in 93% (Monte Carlo standard error 1.5%).

A tactic at true cost 0.05 is scaled in 29% of draws, and a tactic above the cap in at most 1%. The lower-bound rule buys protection against funding a tactic above the cap with missed tactics below it. The misses come from the width of the experiment interval, so the size of the test, not the cap, is what would change them.

Coverage is 93% in every row. The rows reuse the same 300 seeds and plant an additive shift, so the interval moves with the truth and coverage cannot differ. That rate sits 1.4 standard errors from 95%.

What would overturn the scale decision

An experiment lower bound that is not strictly positive, or a cost at that bound that crosses the cap. A model that leaves the experiment interval uses the disagreement sentence and the same bound.

Synthetic dataset generated by this repository. Not the geo-holdout panel and not the weekly revenue series.